

Characterization of oxidized boron-doped carbon fiber anodes for Li-ion batteries by analysis of heat of immersion

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Abstract

Surface oxidation treatment in the temperature range of 500–1400 °C was applied to a graphitized boron-doped mesophase-pitch-based carbon fiber (B-MCF), anode material for Li-ion battery. The correlation between physical properties and lithium intercalation of B-MCF was investigated from measurement of heat of immersion. The surface analysis of heat of immersion and X-ray photoelectron spectroscopy (XPS) indicated that the strength of surface electrostatic field of B-MCF increased with the composition of oxygen on the surface as a result of treatment of the oxidation. Analysis of electrochemical impedance spectroscopy (EIS) indicated that the charge transfer resistance of lithium intercalated B-MCF anode decreased with increase of surface electrostatic field strength. Oxygen atoms on surface as a result of treatment of oxidation caused high surface electrostatic field strength that enhanced an affinity of B-MCF for electrolytes. It was supposed that the low impedance of oxidized B-MCF anode was due to the enhancement of surface polarity and the affinity.

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Keywords: Graphite anode; Surface oxidation; Heat of immersion; XPS; Surface polarity

1. Introduction

High power and high energy density batteries are required for portable electronic products and electric vehicles. Li-ion batteries have attracted great interest due to their high voltage and high energy density, and consequently, research and development of these batteries is in progress worldwide. Artificial graphite and natural graphite have been utilized as anode materials for commercial Li-ion batteries. However, in actual applications, a number of problems associated with the graphite anode remain to be overcome in order to improve high rate capability and cycle ability. Therefore, the performance of graphite anodes has been further improved by a modification of graphite materials, a treatment of carbon surface, use of graphitized carbon fiber and spherical graphite, and doping of hetero-elements [1–7]. We have already reported that the high diffusivity of Li ions in graphitized carbon anode is a key property in enhancing the charge–discharge performance and cycle life [8]. A graphi-

tized mesophase-pitch-based carbon fiber (MCF) with the radial texture in the cross section of the carbon fiber showed excellent charge–discharge characteristics and rapid diffusion of Li ions because Li ions can intercalate smoothly around the carbon surface [8,9]. The performance of the MCF anode has been improved by doping boron in the MCF [10–12]. Boron atom can be substitutionally incorporated into the graphite layer, and its doping leads to the enhancement of the graphitization of carbons. The boron-doped mesophase-pitch-based carbon fiber (B-MCF) exhibited a reversible capacity 10% larger than that of the MCF anode.

Surface oxidation of carbon materials has been investigated to enhance the high rate capability and good cycle life, and the reversible capacity. The mild oxidation treatment of carbon materials in air, oxygen, or other oxidizing substances creates nanochannels or nanopores around the surface and increases the active surface area. Hence, the treatment is expected to give rise to an increase in the high rate capability. We have applied the surface oxidation to the B-MCF anode for further enhancement of high rate performance.

In this study, the correlation between physical properties and the performance of surface oxidized B-MCF anode was

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investigated by the measurement of heat of immersion. The change of the composition of oxygen and electrostatic field on surface of the surface oxidized B-MCF were evaluated.

2. Experimental

The carbon material employed in this study was B-MCF prepared by adding B_4C to MCF and highly graphitized at $3000^\circ C$ in Petoca Material Corporation Limited. The average particle size is $25\ \mu m$ and specific surface area is approximately $0.5\ m^2/g$. Oxidation treatments were carried out in an electric furnace under air or oxygen flow at various heat-treatment temperatures (HTTs). The powder of B-MCF was placed in a magnesia crucible. The time of treatment at various HTTs was arranged so as to gain the yield of about 95%. Thermogravimetry-differential thermal analysis (TG-DTA) curve of B-MCF in air was obtained in the temperature range from the room temperature to $1000^\circ C$ at $10^\circ C/min$ scan speed.

Charge (intercalation)–discharge (deintercalation) cycling tests and the impedance measurements for carbon electrodes were carried out using a three-electrode glass cell. The procedure for constructing B-MCF working electrode was the same as that described previously [8]. Metal lithium foil was used as the counter and the reference electrodes. The electrolyte used was a solution of 1.5 M solution of $LiBF_4$ in a mixture of ethylene carbonate (EC) and γ -butyrolactone (GBL) (1:2 by volume).

The charge–discharge tests for the carbon electrodes in Li/carbon cell were carried out by charging to a 0.01 V versus Li/Li^+ for 10 h with a current density limit of $1\ mA/cm^2$ and discharging to 2 V at $1\ mA/cm^2$.

Electrochemical impedance spectroscopy (EIS) of the lithiated carbon electrodes was carried out in an open-circuit condition in the frequency range from 100 kHz to 0.005 Hz by using an electrochemical interface SI1287 (Solartron) and an impedance/gain-phase analyzer SI1260 (Solartron).

Surface properties of carbon samples were investigated by measurements of XPS, surface area and heat of immersion. XPS spectrum was obtained by using an ESCA-300 (Scienta) with monochromated Al $K\alpha$ radiation. The powder sample was placed in a hollow of aluminum sample holder. Surface composition was calculated from the ratio of peak area of each element using the relative sensitivity factor determined by J.H. Scofield. The specific surface area of each carbon sample was determined with the Brunauer, Emmet, and Teller (BET) method using N_2 -He gas. Measurements of heat of immersion of samples were carried out by a microcalorimeter MMC-5111 (Tokyo Riko Inc.).

Fig. 1 shows a schematic of the calorimeter for the measurement heat of immersion. A solvent is poured into a sample vessel and a reference vessel. Also, a carbon sample is put in a heat-sealed glass ampoule. A sample ampoule and an empty ampoule in each vessel was broken at the same time while stirring the solvent. It is possible to determine

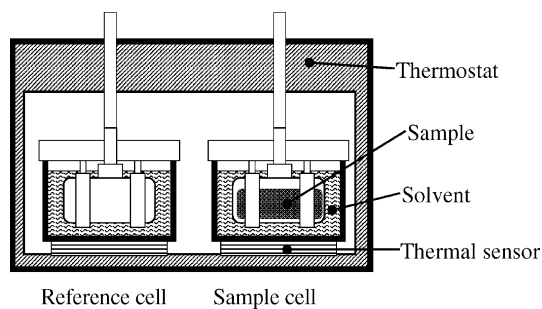


Fig. 1. The apparatus of the micro calorimeter used in this study.

the generated heat by measuring the change of temperature difference between both vessels.

The heat of immersion of samples in heptane and 1-nitropropane was measured for the purpose of analyzing polarity on surface of B-MCF samples with the equations below [13,14]. The immersion heat ΔH_i denotes the sum of the dispersion force h_i^d , the polarizing force h_i^α and the electrostatic force, the interaction between the permanent dipole of the liquid and the static electric field on surface of the solid, h_i^μ .

$$\Delta H_i = h_i^d + h_i^\alpha + h_i^\mu \quad (1)$$

If $h_i^d + h_i^\alpha$ in Eq. (1) is considered to be constant, ΔH_i corresponds to the change in h_i^μ . If F represents the strength of the electrostatic field on the solid surface, the relationship denoted by Eq. (2) given below is established:

$$h_i^\mu = -n\mu F \quad (2)$$

It follows that Eq. (3) given below is established:

$$-\Delta H_i = n\mu F + \text{const} \quad (3)$$

In Eqs. (2) and (3), μ represents the dipole efficiency of the immersion liquid and n the number of adsorbed molecules per unit surface area. If the immersion heat is obtained by using a series of adsorbates differing from each other in the dipole efficiency and being equal to each other in the adsorption area of a single molecule, it is possible to obtain the intensity of the static electric field on the surface, i.e., the surface polarity, from the relationships specified in formulas (1)–(3) given above. *n*-Heptane and 1-nitropropane are equal to each other in the adsorption area of a single molecule. The area occupied per molecule is about $20\ \text{\AA}^2$. Therefore, n comes to approximately 5×10^{18} molecules/ m^2 [13]. The dipole efficiency μ of *N*-heptane and 1-nitropropane is 0 and 3.75 Debye, respectively. It follows that the strength of the electrostatic field on the solid surface F can be determined by the difference in the immersion heat between these two solvents, based on Eq. (3).

Measurements of heat of immersion of each sample were also performed in ethyl carbonate (EC)/ γ -butyrolactone (GBL) 1:2 solvent and EC/methyl ethyl carbonate (MEC) 1:2 solvent. It is possible to make a relative evaluation of the affinity and the wettability of several solid surfaces for

a polar solvent by the heat of immersion [13]. The affinity and wetting of samples in two types of electrolyte solvents were evaluated from the heat of immersion.

3. Results and discussion

3.1. Change of surface properties by the oxidation treatment

A B-MCF employed in this study had radial texture of graphene crystalline in the cross section of the carbon fiber. And it had highly graphitized structure by heating at 3000 °C. It should be noted that the physical and chemical effects of surface oxidation depend strongly on particle shape and arrangement of graphene crystalline since the chemical modification occurs mainly at the prismatic planes [15]. The surface oxidation had a good effect on the B-MCF because its surface consisted mostly of prismatic planes while its highly graphitized inner structure wasn't influenced by heating at below 1400 °C during the oxidation treatment. So, the B-MCF used in this study was extremely suited to the investigation of surface treatment.

The TG profile of B-MCF in air is shown in Fig. 2. Oxidation conditions and specific surface area of each carbon sample are shown in Table 1. It is apparent from TG profile that reactivity of B-MCF with oxygen in air increased drastically beyond about 700 °C. A part of B-MCF was burnt off in the oxidation treatment, generating carbon oxide. The yield is the weight ratio of B-MCF before and after oxidation treatment. It took 12 h at 550 °C, 6 h at 600 °C and 1.5 h at 750 °C to gain the yield of 90–95%. In the case of treatment at 450 °C, weight reduction of B-MCF was only 0.7% despite heating in the oxygen flow. In the temperature range above 750 °C, it was difficult to control the yield, and all parts of sample were evenly oxidized due to the high reactivity. At 1400 °C, sample was put in a closed cup in air and heated for 4 h. As shown in Table 1, the large increase of specific surface area was observed on oxidized B-MCFs. In the temperature range below 750 °C, sample number 4 which was treated at 600 °C had the largest surface area. This result can be explained with TG-DTA curve.

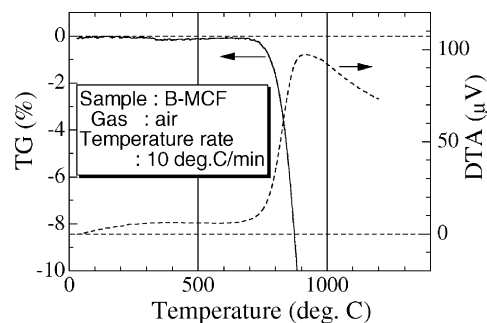


Fig. 2. TG-DTA curves B-MCF in air atmosphere measured at heating rate of 10 °C min⁻¹.

At low temperature, oxidation of graphitized carbon occurs only at the active points such as defects and lowly crystallized parts. On the other hand, at sufficiently high temperature, the oxidation proceeds on the whole part of carbon containing highly graphitized crystal. Judging from TG-DTA curve, the rapid oxidation of graphitized crystal started from about 700 °C on B-MCF used in this study. Therefore, finer structure which led to large specific surface area was grown in the case of 550–650 °C treatment by more selective oxidation, whereas sample number 6 treated at 750 °C had rougher surface structure which led lower to specific surface area. Sample number 7 showed drastically large surface area caused by the strong oxidation treatment at 1400 °C. It is considered that the porous structure was developed in the outer layer of sample number 7 because B-MCF had reached the equilibrium with carbon monoxide at sufficiently high temperature in the limited amount of oxygen.

Fig. 3 shows the XPS spectra of C1s, N1s, O1s and B1s of samples. Intensity of spectra of each sample are normalized. All carbon samples contained four elements, carbon, boron, oxygen and nitrogen. Nitrogen exists on B-MCF as boron nitride because boron at carbon surface reacts with nitrogen in air during the graphitizing process. The spectrum of each element was divided into some peaks by its binding state. In C1s spectrum, in additions to main large C–C bonding peak, two peaks assigned to B–C (282.5 eV) and C–O (286.5 eV) bonding were observed. There was no difference in peak of B–C bonding among all samples and it is suggested that B–C bonding existed mainly in boron-graphite solid solution.

Table 1
Surface area and elemental composition dependence on oxidation treatment

Sample number	Condition of oxidation treatment	Yield (%)	Specific surface area (m ² /g)	Surface composition (atomic ratio)			
				C	O/C	B/C	N/C
1	Untreated	–	0.48	1	0.020	0.02	0.01
2	450 °C/9 h oxygen flow	99.3	0.69	1	0.009	0.02	0.01
3	550 °C/12 h open air	93.8	5.25	1	0.025	0.02	0.02
4	600 °C/3 h open air	93.7	5.95	1	0.042	0.03	0.02
5	650 °C/3 h open air	93.1	5.53	1	0.038	0.03	0.02
6	750 °C/1 h open air	93.4	3.26	1	0.051	0.04	0.03
7	1400 °C/4 h air (closed cup)	80.1	12.4	1	0.022	0.01	0

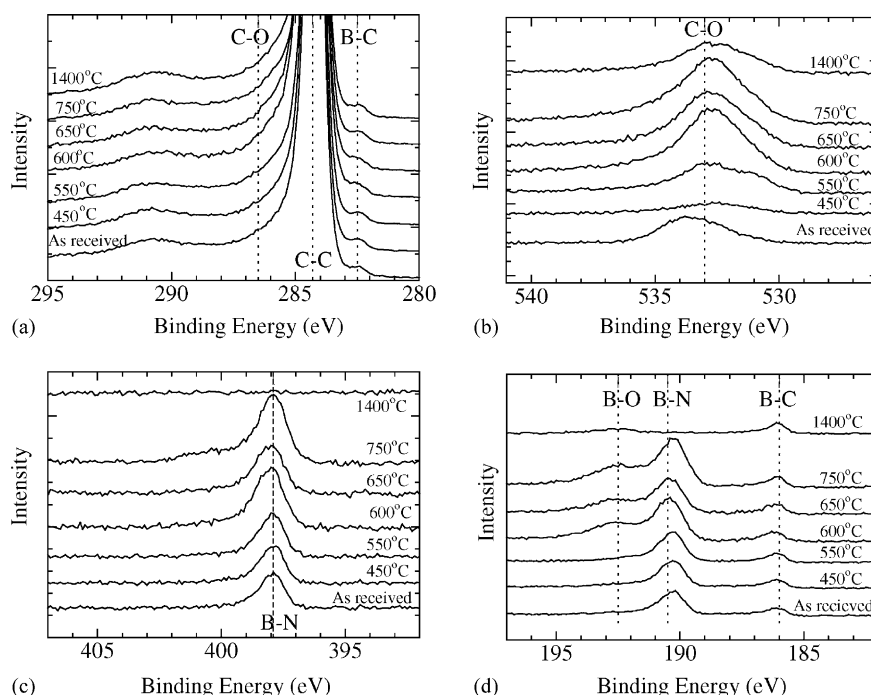


Fig. 3. (a) C1s, (b) O1s, (c) N1s and (d) B1s XPS spectra of normal and oxidized B-MCF samples. Dotted lines in figure indicate peak positions of each bonding.

In O1s spectrum, the peak centered at 532–533 eV was assigned to C–O and B–O bonding. The peak intensity increased with oxidation temperature below 750 °C and oxygen content of samples oxidized at 600–750 °C was much larger than for other samples. O1s peak position of initial untreated B-MCF was shifted from peaks of oxidized B-MCFs. It is considered that the initial binding state of oxygen on surface of untreated B-MCF was different from those of oxidized B-MCFs.

In N1s spectrum, only one peak at 398 eV assigned to B–N bonding was observed. In B1s spectrum, peaks were assigned to B–C (186 eV), B–N (190.5 eV) and B–O (192.5 eV) bonding. B–N peak of each sample showed the same change in N1s and B1s spectrum. B–N bonding peak was magnified slightly from initial state to 750 °C oxidation and disappeared in spectrum of sample number 7. Intensity of B–O peak was influenced by the oxidation condition. B–O peak appeared in the spectra of samples oxidized at

more than 600 °C and it is considered that B–O compounds such as B_2O_3 were generated on B-MCF by oxidation at 600–1400 °C. On the other hand, B–C bonding was not affected by oxidation.

Table 1 shows the atomic composition on surface of carbon sample evaluated from XPS shown in Fig. 3. Boron and nitrogen contents increased slightly on sample numbers 2–6 because boron nitride did not react under this condition and only carbon was removed from surface by oxidation. At 1400 °C, the composition of B and N was almost zero since boron nitride vaporized. Consequently, sample number 7 had extremely large specific surface area. The content of oxygen composite increased with heating temperature of sample numbers 2–6. In oxidation process, oxygen atoms are introduced to active site on carbon surface and then dissociated from carbon as CO [16]. A part of oxygen atoms introduced on B-MCF remained on surface. Hence, larger oxygen content was observed on sample number 6 because

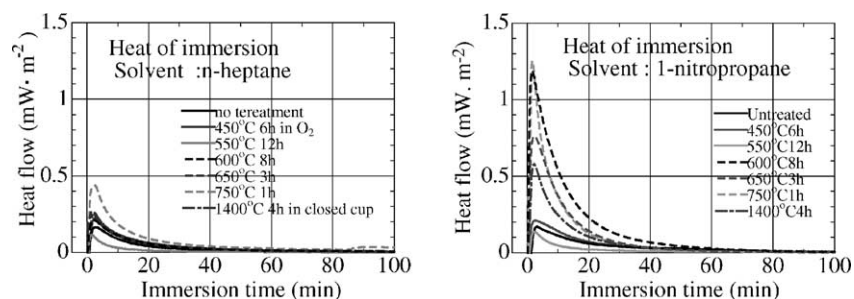


Fig. 4. Thermogram for the immersion heat of oxidized B-MCF samples with *n*-heptane and 1-nitropropane.

B-MCF was in more activated state in the case of treatment at higher temperature. Judging from the analysis of XPS spectrum, oxygen atoms were bonded with carbon and boron on surface of B-MCF. It is considered that low surface oxygen content of sample number 7 was affected by the absence of boron.

The chart presented in Fig. 4 shows heat generating rate against time during immersion. In Fig. 4, the area of chart indicates the amount of generated heat and the sharpness of heat generation peak is related to wetting between solvent and solid. The immersion heat was calculated by integrating heat generation peak in Fig. 4. The calculation of the strength of the surface electrostatic field was done by Eq. (3) with following parameters:

- $-\Delta H_i$: heat of immersion to *n*-heptane and 1-nitropropane (J/m^2);
- μ : *n*-heptane: $\mu = 0$ Debye, 1-nitropropane: $\mu = 3.75$ Debye (1 Debye = $3.33564 \times 10^{-30} \text{ C m}$); and
- n : $n = 5 \times 10^{18}$ molecules/ m^2 .

The calculated results for immersion heat and strength of electrostatic field are shown in Table 2. Immersion heat of *n*-heptane represents heat of physical adsorption on carbon sample and difference of immersion heat between *n*-heptane and 1-nitropropane related to the interaction between a polarity of solvent and surface of B-MCF.

Immersion heat caused by physical adsorption per unit surface area should be equivalent, but remarkable differences of immersion heat per surface area in *n*-heptane was observed among the samples. As described above, these samples, and especially those treated at low temperature, had fine structures on surface, and thus *n*-heptane was not adsorbed on the entire BET surface area. It is not considered that this affected evaluation of the average strength of electrostatic field because *n*-heptane and 1-nitropropane are almost equal in molecule size.

In Fig. 5, the calculated strength of electrostatic field is plotted against surface oxygen content. The strength of electrostatic field indicated a tendency to increase basically with oxygen composition. One thousand four hundred and 750 °C samples are out of the line containing other samples. These samples experienced strong oxidation and their surface structure such as roughness, a form of oxygen function

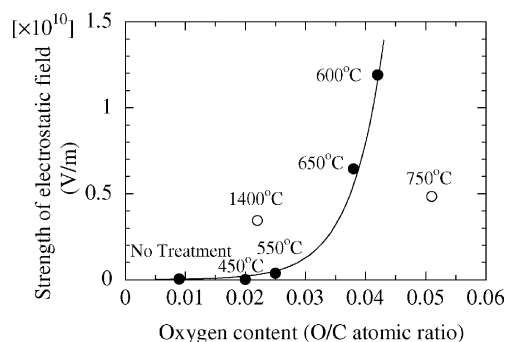


Fig. 5. Correlation between strength of surface electrostatic field and oxygen content of oxidized B-MCF samples.

group and distribution of boron nitride and boron oxide was very different from other samples heated at lower temperature. So, sample numbers 6 and 7 showed a different relationship for the oxygen content. It is assumed that the polarity on the surface of the carbon was generated by oxygen composite and the functional group containing the oxygen atoms. On surface of sample oxidized at 600–750 °C, where XPS spectrum had large B–O bonding peak, especially large strength of surface electrostatic field was observed.

Table 3 show the results of measurements of immersion heat of carbon samples in EC + GBL and EC + MEC solvents. Sample number 5 shows the largest immersion heat per unit area and the best wetting in each solvent.

Heat of immersion of carbon samples per unit area to electrolyte solvent is plotted against the strength of surface electrostatic field in Fig. 6. Heat of immersion to EC + GBL solvent was almost proportional to surface electrostatic field strength. On the other hand, in EC + MEC solvent, the relationship between immersion heat and strength of surface electrostatic field was not clear. These results can be explained in terms of the interaction of EC + GBL solvent with electrostatic field on B-MCF due to higher polarity of GBL [17]. In the case of polar solvent, the immersion heat is considered to represent the affinity because a dipole–dipole interaction accounts for a large part of the affinity if a solid and a solvent are polar. Thus, the affinity was proportional to surface electrostatic field in EC + GBL solvent. It becomes evident that the affinity between B-MCF and a non-aqueous electrolyte composed of polar solvents and salts can

Table 2

Immersion heat of oxidized B-MCF samples to *n*-heptane and 1-nitropropane and strength of surface electrostatic field

Sample number	Oxidizing condition	Heat of immersion (mJ/g)		Heat of immersion (mJ/m^2)		Surface electrostatic field strength (V/m)
		<i>n</i> -Heptane	1-Nitropropane	<i>n</i> -Heptane	1-Nitropropane	
1	No treatment	213.8	110.1	445.4	229.4	–
2	450 °C/9 h in oxygen flow	187.8	189.1	272.2	274.0	$3.10\text{E}+07$
3	550 °C/12 h in air	397.8	511.9	75.8	97.5	$3.65\text{E}+08$
4	600 °C/8 h in air	1298.5	5510.6	218.2	926.1	$1.19\text{E}+10$
5	650 °C/3 h in air	1240.4	3361.0	224.3	007.8	$6.44\text{E}+09$
6	750 °C/1 h in air	1117.2	2055.2	342.7	630.4	$4.83\text{E}+09$
7	1400 °C/4 h in closed crucible	3055.4	5588.8	246.4	450.7	$3.43\text{E}+09$

Table 3

Immersion heat for the immersion of oxidized B-MCF samples with EC + GBL and EC + MEC

Sample number	Oxidizing condition	EC + BL		EC + MEC	
		mJ/g	mJ/m ²	mJ/g	mJ/m ²
1	No treatment	51.7	107.7	69.1	144.0
2	450 °C/9 h in oxygen flow	71.8	104.1	122.1	176.9
3	550 °C/12 h in air	610.8	116.3	566.6	107.9
4	600 °C/8 h in air	2121.1	356.5	1297.5	218.0
5	650 °C/3 h in air	1206.1	218.1	1471.5	266.1
6	750 °C/1 h in air	776.3	238.1	877.1	269.1
7	1400 °C/4 h in closed crucible	2416.5	194.9	1838.3	148.3

be improved by increasing the polarity on the surface of the B-MCF.

3.2. Anode properties of oxidized B-MCF

Charge and discharge capacities and coulomb efficiency at the first cycle are shown in Table 4. The increase of the discharge capacity of B-MCF was 3–5% due to the oxidation treatment. The B-MCF has lowly graphitized carbon layer on its surface. These results concerning reversible capacity are estimated theoretically as low crystallinity part having lower lithium storage capacity was removed during oxidation treatment.

The coulomb efficiency at the first cycle did not change in spite of increasing specific surface area, except in the case of the carbon samples at 750 and 1400 °C. These samples had a higher ratio of graphene edge surface, which was active in the electrolyte decomposition, than other samples due to strong oxidation condition under which highly graphitized crystal oxidized rapidly.

Fig. 7 shows the impedance spectra of electrode of lithiated B-MCF. It is reported that the impedance spectrum of lithium intercalated graphite anode exhibited two semicircles with the inclined line [18,19]. The impedance spectra of oxidized B-MCFs are smaller than that of untreated carbon sample. In the Nyquist plot of B-MCF samples in Fig. 7, two or three separate semicircles are observed. Fig. 8 shows a series of impedance spectra measured at the different equilib-

rium potentials during the first discharge (de-intercalation). The high frequency part containing the first semicircle is independent of anode potential, whereas semicircles in the middle frequency part expanded and time constant changed with anode potential.

The semicircles are attributed to lithium migration through the surface protective film on B-MCF and charge transfer process. The resistance of the surface protective film is not influenced by anode potential. Charge transfer resistance is dependent on electrode potential because Li intercalated graphite undergoes phase transition between various intercalation stages during charge and discharge process [19]. Hence, the semicircle in the high frequency

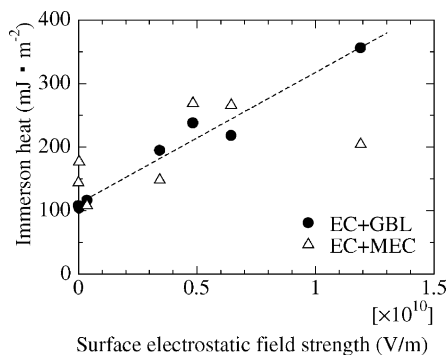


Fig. 6. Immersion heat of oxidized B-MCF samples to EC + GBL and EC + MEC as a function of the strength of surface electrostatic field.

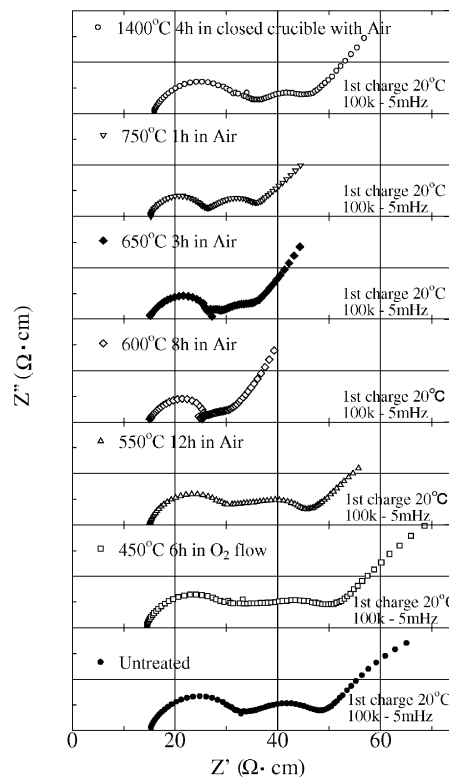


Fig. 7. Nyquist plots for normal and oxidized B-MCF samples lithiated at constant current of 4 mA and at constant potential at 10 mV vs. Li/Li⁺ for 10 h in 1.5 M LiBF₄/EC + GBL (1:2). These measurements were carried out a day after charging to stabilize surface state of the electrodes.

Table 4

Charge–discharge characteristics of oxidized B-MCF samples in 1.5 M LiBF₄/EC + GBL (1:2)

Sample number	Condition of heat treatment	1st charge capacity (mAh/g, 0.01 V)	Reversible capacity (mAh/g, 0.01–2 V)	Initial Coulomb efficiency	Oxidation yield (%)
1	Untreated	374	337	90.1	–
2	450 °C/9 h O ₂ flow	377	341	90.4	99.3
3	550 °C/12 h open air	383	345	90.1	93.8
4	600 °C/8 h open air	384	347	90.4	93.7
5	650 °C/3 h open air	388	346	89.2	93.1
6	750 °C/1 h open air	400	352	88.1	92.5
7	1400 °C/4 h air (closed crucible)	412	350	85.0	80.1

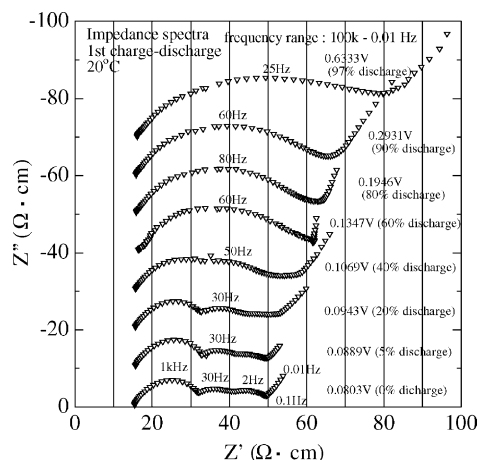


Fig. 8. A series of Nyquist plots of normal B-MCF measured at different equilibrium potentials during 1st discharge.

region is assigned to resistance of Li-ion migration through surface film interface on B-MCF. It is considered that charge transfer resistance caused other semicircles in the middle frequency region. It is supposed that the inclined line is impedance of lithium diffusion within B-MCF.

To evaluate the influence of surface oxidation on electrochemical properties of B-MCF anode, the measured impedance spectra analyzed using the equivalent circuit shown in Fig. 9 and the values of parameters were determined by modeling calculation. The equivalent circuit contains a series circuit of three components, Randles circuit for lithium intercalation reaction, a parallel circuit of resistance and capacitance for lithium migration through surface film and a resistance for a resistance of electrolyte. Similar

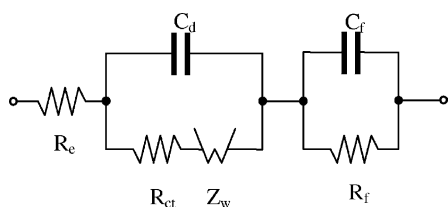


Fig. 9. Equivalent circuit used for the analysis of EIS data. Re: resistance of electrolyte; C_f and R_f: capacitance and resistance of surface film; C_d: double layer capacity; R_{ct}: charge transfer resistance; Z_w: Warburg impedance.

type of this circuit has been applied usefully to analyze the impedance spectra of graphite anode [19,20], although the constant phase element should be placed instead of C_{dl} in Randles circuit allowing for a factor of porous structure of electrode in the strict sense [18]. In this study, the simple equivalent circuit shown in Fig. 9 was used in order to simplify impedance factors and compare calculated parameters with the experimental physical data.

The calculated results are shown in Table 5. Semicircles in the higher and the lower frequency regions are assigned to film resistance and charge transfer resistance of B-MCFs, respectively. Therefore, C_f and R_f in Table 5 are related with the film resistance. C_d and R_{ct} represent properties of the charge transfer resistance. In Fig. 10, C_fR_f and C_dR_{ct} are plotted versus surface properties, surface electrostatic field strength, and immersion heat to EC + GBL solution, oxygen content and specific surface area of BMCF.

C_d and R_{ct} tended to change monotonously with surface electrostatic field strength, immersion heat and oxygen content. There was no correlation between C_d, R_{ct} and specific surface area. C_d increased proportionally with heat of immersion. It is supposed that heat of immersion represents affinity between solid surface and solvent. Therefore, the linear increase of C_d with immersion heat of EC + GBL solvent suggests that adsorption of electrolyte containing Li salt in the electric double layer on B-MCF was also enhanced by the improved affinity between carbon surface and solvent. R_{ct} decreased also proportionally with strength of surface electrostatic field. This means that the reaction resistance of Li intercalation reaction became smaller with increasing

Table 5

Results of fitting calculation of impedance spectra for equivalent circuit and surface

Sample number	HTT (°C)	Parameter for the equivalent circuit			
		R _e (Ω cm ²)	C _d (F/cm ²)	R _{ct} (Ω cm ²)	Z _w
1	–	16.16	0.00153	12.8	0.203
2	450	15.39	0.00084	14.3	0.176
3	550	15.81	0.00120	12.9	0.297
4	600	14.59	0.00783	3.34	0.309
5	650	15.79	0.00375	5.34	0.282
6	750	10.81	0.00752	7.76	0.359
7	1400	16.92	0.00252	9.11	0.250

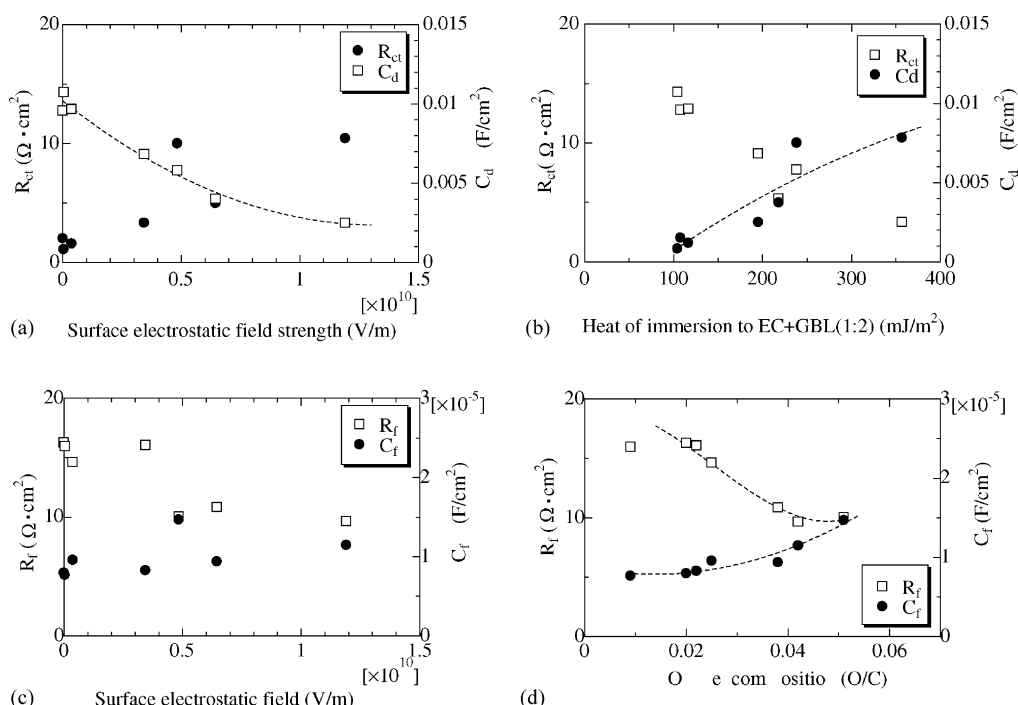


Fig. 10. Correlations between circuit elements and surface physical properties of B-MCF samples. (a) R_{ct} and C_d vs. surface polarity of B-MCF. (b) R_{ct} and C_d vs. affinity to electrolyte. (c) R_f and C_f vs. surface polarity. (d) R_f and C_f vs. surface oxygen content.

surface polarity of B-MCF. It is assumed that polar surface could attract Li ions and reduce the activation energy for Li intercalation reaction functioning as a catalyst.

It is supposed that C_f and R_f are related to Li migration resistance in protective film on B-MCF. C_f and R_f also showed the same tendency as C_d and R_{ct} regarding the surface properties of B-MCF, although they were less effected than C_d and R_{ct} . C_f and R_f indicated the most obvious relationship with oxygen content on surface of B-MCF. Parameters of film resistance are dependent on thickness and uniformity of film. It is well known that surface film formation and its properties are highly dependent on the composition of the electrolyte solvents [21]. Thus, the chemical reactivity between electrolyte components and active material is thought to be a major factor in protective film formation. It is estimated that the film forming reaction was also influenced by oxygen content on B-MCF surface and film properties were improved by the increase of surface oxygen content.

Although B-MCF was used as the raw material in this study, the oxidation treatment will work effectively for other carbon materials not doped with boron. However, judging from the results of XPS measurement, it is evident that boron–oxygen compounds such as B_2O_3 were formed by proper oxidation treatment of boron-doped carbon and increased oxygen content on its surface, which led to larger affinity and surface polarity. Therefore, it is estimated that in the case of boron-doped graphitized carbon surface oxidation treatment is more effective for improving anode properties than in the case of normal graphitized carbon.

It is reported that electrochemical properties of graphite anode can be improved by coating with oxides such as Al_2O_3 , ZrO_2 and LiPdO_3 and coated graphite anodes had much smaller charge transfer resistance in EIS measurement [22–24]. It is supposed that these oxides enhance surface polarity and affinity with electrolyte. The B–O compounds on oxidized B-MCF surface played the same role as these oxides and made surface oxidation more effective. Hence, the surface oxidation treatment is especially suited to boron-doped graphitized carbon.

4. Conclusion

In this paper, the correlation between surface properties and the electrochemical properties of oxidized B-MCF anode of the lithium ion battery has been investigated. The measurement of heat of immersion, XPS and BET surface area were employed to investigate the surface state of oxidized B-MCF. Anode properties of oxidized B-MCF were evaluated with the electrochemical impedance spectroscopy.

From the results of XPS and BET measurement, depending on temperature of the treatment, specific surface area and oxygen content on B-MCF increased. Immersion heat analysis revealed that the strength of surface electrostatic field on B-MCF has a tendency to increase with increasing the content of surface oxygen. The affinity to EC + GBL electrolyte also increases with strength of surface electrostatic field. In EIS, the charge transfer resistance of lithium ion intercalation reaction on B-MCF was reduced proportionally

to strength of surface electrostatic field. These results suggest that stabilized interface between electrolyte and surface of B-MCF due to enhanced surface polarity of B-MCF leads to lower resistance of lithium intercalation reaction. On the other hand, the lithium migration resistance through surface protective film also decreases with increase of surface oxygen composition. It is considered that thinner, more homogeneous protective film with less lithium ionic resistance can be formed on oxidized B-MCF which has larger oxygen content.

Analysis of heat of immersion is useful for investigating surface state and interfacial properties between carbon material and electrolyte. It should be noted that the surface polarity and the affinity for electrolyte are among the most important properties for carbon electrode materials in the lithium ion battery. Further studies are needed in order to understand and explain the detailed mechanisms whereby resistance is reduced.

References

- [1] K. Kikuchi, Y. Ikezawa, T. Takamura, J. Electroanal. Chem. 396 (1995) 451.
- [2] M. Yoshino, H. Wang, K. Fukuda, Y. Hara, Y. Adachi, J. Electrochem. Soc. 147 (2000) 1245.
- [3] E. Peled, C. Menachem, D. Bar-Tow, A. Melman, J. Electrochem. Soc. 143 (1996) L4.
- [4] K. Suzuki, T. Hamada, T. Sugiura, J. Electrochem. Soc. 146 (1999) 890.
- [5] M. Hara, A. Satoh, N. Takami, T. Ohsaki, Tanso 165 (1994) 261.
- [6] Y. Ein-Eli, V.R. Koch, J. Electrochem. Soc. 144 (1997) 2968.
- [7] M. Endo, C. Kim, T. Karai, T. Tamaki, Y. Nishimura, M.J. Matthews, S.D.M. Brown, Phys. Rev. B 58 (1998) 8991.
- [8] N. Takami, A. Satoh, M. Hara, T. Ohsaki, J. Electrochem. Soc. 142 (1995) 2564.
- [9] T. Ohsaki, N. Takami, A. Satoh, T. Kishi, K. Sakurai, M. Kanda, in: Proceedings of the Symposium on Lithium Polymer Batteries, Electrochemical Society, 1997, p. 265.
- [10] Y. Nishimura, T. Takahashi, T. Tamaki, M. Endo, M.S. Dresselhaus, Tanso 172 (1996) 725.
- [11] C. Kim, T. Fujino, K. Miyashita, T. Hayashi, M. Endo, M.S. Dresselhaus, J. Electrochem. Soc. (2000) 1257.
- [12] B.M. Way, J.R. Dahn, J. Electrochem. Soc. 141 (1994) 907.
- [13] J.J. Chessick, A.C. Zettlemoyer, Adv. Catal. 6 (1959) 263.
- [14] S. Hagiwara, K. Tsutsumi, H. Takahashi, Carbon 9 (1971) 693.
- [15] R.I.R. Blyth, H. Buqa, F.P. Netzer, M.G. Ramsey, J.O. Besenhard, P. Golob, M. Winter, Appl. Surf. Sci. 167 (2000) 99.
- [16] N.R. Laine, F.J. Vastr, P.L. Walker Jr., in: Proceedings of the 5th Conference on Carbon, Buffalo, 1961, p. 211.
- [17] J.-P. Gabano, Lithium Batteries, Academic Press, 1983, p. 16.
- [18] A. Funabiki, M. Inaba, Z. Ogumi, J. Power Sources 68 (1997) 227.
- [19] D. Aurbach, B. Mrkovsky, I. Weissman, E. Levi, Y. Ein-Eli, Electrochim. Acta 45 (1999) 67.
- [20] R. Yazami, in: G. Pistoia (Ed.), Lithium Batteries, New Materials, Developments and Perspectives, Elsevier, Amsterdam, London, New York, Tokyo, 1994, p. 88.
- [21] R. Yazami, Electrochim. Acta 45 (1999) 87.
- [22] S. Kim, Y. Kodama, H. Ikuta, Y. Uchimoto, M. Wakihara, Electrochem. Solid State Lett. 4 (2001) A109.
- [23] I.R.M. Kottogoda, Y. Kodama, H. Ikuta, Y. Uchimoto, M. Wakihara, Electrochem. Solid State Lett. 5 (2002) A275.
- [24] P. Yu, B.S. Haran, J.A. Ritter, R.E. White, B.N. Popov, J. Power Sources 91 (2000) 107.